

1. Display the count of the number of rows in the Animal table.

```
Select NoAnimals=count(*) from Animal
```

) %	
Results	Messages
NoAnimals	
26	

2. Delete a specified record in a table and prove that it is deleted. (before and after)
I will be deleting record 28 which is specified as bedroom

Before:

SQL Server Enterprise Edition (64-bit) - Habbitat

```
Select * from Habbitat
where HabbitID = 28
```

100 %

Results Messages

	HabbitID	Name	Desc
1	28	Bedroom	NULL

After:

```
Select * from Habbitat
where HabbitID = 28

Delete from Habbitat
where HabbitID = 28

Select * from Habbitat
where HabbitID = 28
```

100 %

Results Messages

(1 row affected)

(1 row affected)

(0 rows affected)

Completion time: 2025-06-15T13:30:07.2999970-04:00

```
[-] Select * from Habitat
    where HabitatID = 28
[-] Delete from Habitat
    where HabitatID = 28
[-] Select * from Habitat
    where HabitatID = 28
```

100 %

Results Messages

HabitatID	Name	Desc
-----------	------	------

3. Two table join qualified with a key value.

```
[-] Select animal.Name, class.Name
    From Animal, Class
    Where animal.ClassID=animal.ClassID
```

100 %

Results Messages

	Name	Name
1	Black Bear	Mammal
2	Rattlesnake	Mammal
3	Coyote	Mammal
4	Whale	Mammal
5	Walrus	Mammal
6	Polar Bear	Mammal
7	Clown Fish	Mammal
8	Octopus	Mammal
9	Dolphin	Mammal
10	Shark	Mammal
11	Barred Owl	Mammal
12	Horned Owl	Mammal

4. Three table join qualified with a key value.

```
Select A.Name, R.Name
From Animal A, Range R, AnimalRange AR
Where A.AnimalID = AR.AnimalID
and AR.RangeID = R.RangeID
```

100 %

Results Messages

	Name	Name
1	Black Bear	Temperate Forest
2	Black Bear	Wetlands
3	Rattlesnake	Desert
4	Rattlesnake	Temperate Forest
5	Coyote	Desert
6	Coyote	Temperate Forest
7	Coyote	Mountain Range
8	Coyote	Wetlands
9	Coyote	Coastal
10	Whale	Ocean
11	Walrus	Ocean
12	Walrus	Arctic
13	Polar Bear	Arctic
14	Clown Fish	Ocean
15	Octopus	Ocean
16	Dolphin	Ocean
17	Shark	Ocean
18	Barred Owl	Temperate Forest
19	Homed Owl	Temperate Forest
20	Screech Owl	Temperate Forest
21	Barn Owl	Temperate Forest
22	Timber Wolf	Temperate Forest

5. Use twelve SQL functions (such as SUM). This should be twelve queries with at least one function for each query.

i.

```
select Noanimals = count(*) from animal
```

100 %	
Results	Messages
Noanimals	
1	26

ii.

```
Select MaxName=Max(Name) from animal
```

100 %	
Results	Messages
MaxName	
1	Whale

iii.

```
Select MinName=Min(Name) from animal
```

100 %	
Results	Messages
MinName	
1	Bam Owl

iv.

```
Select MinID=Min(AnimalID) from animal
```

100 %	
Results	Messages
MinID	
1	1

v.

```
Select SystemDate=GETDATE();
```

Results	
Messages	
SystemDate	
1	2025-06-15 13:49:46.867

vi.

```
SELECT currenttimestamp=CURRENT_TIMESTAMP;
```

Results	
Messages	
currenttimestamp	
1	2025-06-15 13:52:14.707

vii.

```
SELECT DATEADD(year, 1, '2017/08/25') AS DateAdd;
```

Results	
Messages	
DateAdd	
1	2018-08-25 00:00:00.000

viii.

```
SELECT DATEDIFF(year, '2017/08/25', '2011/08/25') AS DateDiff;
```

Results	
Messages	
DateDiff	
1	-6

ix.

```
SELECT DATEFROMPARTS(2018, 10, 31) AS DateFromParts;
```

Results	
Messages	
DateFromParts	
1	2018-10-31

x.

```
SELECT DATENAME(year, '2017/08/25') AS DatePartString;
```

100 %

Results Messages

	DatePartString
1	2017

xi.

```
SELECT DAY('2017/08/25') AS DayOfMonth;
```

100 %

Results Messages

	DayOfMonth
1	25

xii.

```
SELECT MONTH('2017/08/25') AS Month;
```

100 %

Results Messages

	Month
1	8

6. Demonstrate the use of the rowcount feature to return 4 rows from a table that has more than 4 rows. Then demonstrate the use of the rowcount feature to reset the number of rows returned to all of the rows in a table.

```
-- animal table actually has 26 rows
set rowcount 4
select * from animal
```

100 %

Results Messages

	AnimalID	Name	Desc	ClassID	HabitatID
1	1	Black Bear	NULL	1	1
2	2	Rattlesnake	NULL	2	3
3	3	Coyote	NULL	1	1
4	4	Whale	NULL	1	1

To reset the rowcount feature one must open a new query like in the screenshot below:

Overlord2404\SQL...imal3 - dbo.Animal SQLQuery3.sql - OV...RD2404\daniy (58))* SQLQuery2.sql - OV...RD2404\daniy (60))*

```
select * from animal
```

100 %

Results Messages

	AnimalID	Name	Desc	ClassID	HabitatID
1	1	Black Bear	NULL	1	1
2	2	Rattlesnake	NULL	2	3
3	3	Coyote	NULL	1	1
4	4	Whale	NULL	1	1
5	5	Walrus	NULL	1	1
6	6	Polar Bear	NULL	1	27

7. Create a SQL routine to insert values into a row.

```
insert into color
(ColorID, ColorName, [Desc])
Values
(10, 'Blue', 'Navy')
```

	ColorID	ColorName	Desc
▶	1	White	NULL
	2	Black	NULL
	3	Brown	NULL
	4	Red	NULL
	5	Grey	NULL
	6	Yellow	NULL
	7	Orange	NULL
	8	Purple	NULL
	9	Green	NULL
	10	Blue	Navy
*	NULL	NULL	NULL

8. Create a SQL routine to update values in a table.

```
update Color
Set [Desc] = 'verydark'
where ColorID = 11
```

100 %

Messages

(1 row affected)

Completion time: 2025-06-15T14:20:42.6983842-04:00

9. Demonstrate the use of a truncate statement for a table.

```
truncate table Animal
```

100 %

Messages

Commands completed successfully.

Completion time: 2025-06-15T14:21:50.6402145-04:00

10. Create a query that uses Group By

```
select C.colormame, animalcount=count(A.name)
from Animal A, AnimalColor AC, Color C
where A.AnimalID = AC.AnimalID
and AC.ColorID = C.ColorID
--and C.ColorName = 'black'
group by c.ColorName
```

100 %

Results Messages

	colomame	animalcount
1	Black	1
2	Brown	8
3	Green	1
4	Grey	5
5	Orange	1
6	Purple	1
7	White	3

11. Demonstrate the use of distinct, ascending, descending in a query.

```
select distinct name from Animal
where name like '%owl'
```

100 %

Results Messages

	name
1	Bam Owl
2	Barred Owl
3	Horned Owl
4	Screech Owl

Ascending order

```
select name from Animal
order by name asc
```

100 %

Results Messages

	name
1	Bam Owl
2	Barred Owl
3	Black Bear
4	Buffalo
5	Cheetah
6	Chicken
7	Clown Fish
8	Cow
9	Coyote
10	Dolphin
11	Fox
12	Green Frog
13	Grey Wolf
14	Horned Owl
15	Kermit Frog
16	Octopus
17	Ostrich
18	Pig
19	Polar Bear
20	Rattlesnake
21	Screech Owl
22	Shark
23	Timber Wolf
24	Turkey
25	Walrus
26	Whale

Descending order

```
select name from Animal
order by name desc
```

100 %

Results Messages

	name
1	Whale
2	Walrus
3	Turkey
4	Timber Wolf
5	Shark
6	Screech Owl
7	Rattlesnake
8	Polar Bear
9	Pig
10	Ostrich
11	Octopus
12	Kermit Frog
13	Horned Owl
14	Grey Wolf
15	Green Frog
16	Fox
17	Dolphin
18	Coyote
19	Cow
20	Clown Fish
21	Chicken
22	Cheetah
23	Buffalo
24	Black Bear
25	Barred Owl
26	Bam Owl

12. Demonstrate @@version

```
select @@VERSION
```

(No column name)	
1	Microsoft SQL Server 2022 (RTM) - 16.0.10006 (X...

13. How many animals were black (number, not the list of animals).

```
select C.colourname, animalcount=count(A.name)
from Animal A, AnimalColor AC, Color C
where A.AnimalID = AC.AnimalID
and AC.ColorID = C.ColorID
and C.ColorName = 'black'
group by c.ColorName
```

	colourname	animalcount
1	Black	1

14. Which color had the most animals.

```
SELECT TOP 1 c.ColorName, COUNT(ac.AnimalID) AS AnimalCount
FROM AnimalColor ac
JOIN Color c ON ac.ColorID = c.ColorID
GROUP BY c.ColorName
ORDER BY COUNT(ac.AnimalID) DESC;
```

	ColorName	AnimalCount
1	Brown	8

15. Which animal was in the most ranges.

```
SELECT TOP 1 a.Name, COUNT(ar.RangeID) AS RangeCount
FROM AnimalRange ar
JOIN Animal a ON ar.AnimalID = a.AnimalID
GROUP BY a.Name
ORDER BY COUNT(ar.RangeID) DESC;
```

100 %

Results Messages

	Name	RangeCount
1	Coyote	5